

Mustaqim Nurul Huda

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PROFILE SUMMARY

Industrial Electrical Engineering graduate from PENS (2026) with a strong background in automation, embedded control systems, and electrical utility maintenance. Experienced in hardware-software integration and executing high-safety electrical diagnostics (K3/OHS & LOTO) in large-scale manufacturing facilities. Innovative problem-solver with a proven track record validated by registered Copyrights and Patent applications. Focused on advancing a career in Automation, Maintenance, or Continuous Improvement Engineering roles within the manufacturing sector.

EDUCATION

Electronic Engineering Polytechnic Institute of Surabaya (PENS) Surabaya, Indonesia
Bachelor of Applied Science (B.A.Sc.) in Industrial Electrical Engineering | **GPA: 3.48 / 4.00** Aug. 2022 – Jul. 2026

- **Relevant Coursework:** Power Electronics, Industrial Automation & Instrumentation, SCADA Systems, Microcontrollers & Embedded Control, Intelligent Control, Power System Protection & Analysis, Smart Grid, Electric Motor Drives, Energy Management & Audit, Occupational Health and Safety (K3/OHS).

FINAL PROJECT

Design and Implementation of MPPT Based on Hybrid WODE Algorithm Surabaya, Indonesia
Electronic Engineering Polytechnic Institute of Surabaya (PENS) Feb. 2024 – Jul. 2026

- **Tools & Technologies:** C++, STM32 Microcontroller, Python (PyQt6), MATLAB/Simulink, Modified Ćuk Converter.
- Engineered and implemented a standalone Maximum Power Point Tracking (MPPT) controller utilizing a Modified Ćuk DC-DC converter topology to directly drive a DC submersible water pump system[cite: 1, 2].
- Developed firmware based on a hybrid Whale Optimization Algorithm and Differential Evolution (WODE) on an STM32 microcontroller to track the Global Maximum Power Point (GMPP) under partial shading conditions (PSC)[cite: 1, 2].
- Achieved an average tracking efficiency of 98.53% with a convergence speed of 3.67 seconds and a steady-state power ripple of 2.28 W validated across 86,790 data samples[cite: 1, 2].
- Constructed a real-time serial GUI monitoring dashboard in Python (PyQt6) for data logging and graphical visualization of voltage, current, power, and system efficiency[cite: 1, 2].

WORK & RESEARCH EXPERIENCE

Electrical Engineering Intern Probolinggo, Indonesia
PT Eratex Djaja Tbk Mar. 2025 – Jun. 2025

- Analyzed the operational performance of a 1.37 MWp On-Grid Solar Power Plant (PLTS) using Solar-Log, ensuring 13.6% factory energy coverage.
- Examined energy management parameters, verifying Power Factor stability ($\cos \varphi = 0.967$) and evaluating solar plant Specific Yield at 3.543 kWh/kWp/day.
- Conducted preventive maintenance on Sungrow SG110CX inverters, DC/AC protection panels, Canadian Solar PV modules, transformer oil purification (vacuum dehydration), and Dielectric Breakdown Voltage (BDV) testing.
- Executed advanced electrical diagnostics, including Megger insulation testing, no-load startup current checks on industrial induction motors, and PV string I-V curve analysis while strictly adhering to LOTO protocols.

Project Lead – PKM-KC Research Grantee Surabaya, Indonesia
Ministry of Education, Culture, Research, and Technology May 2024 – Nov. 2024

- Led a 4-member multidisciplinary team in designing and developing "EcoMow," an eco-friendly zero-emission battery-powered lawn mower funded by a national PKM-KC research grant.
- Engineered a high-efficiency electric propulsion system and battery management strategy to replace traditional internal combustion engines, reducing carbon emissions and noise levels.
- Converted technical manuals and structural designs into an officially registered Intellectual Property Copyright with the Ministry of Law and Human Rights Indonesia (HAKI No. EC00202499819).

ORGANIZATIONAL EXPERIENCE

HIMA ELIN (Industrial Electrical Engineering Student Association) <i>Active Member</i>	Surabaya, Indonesia <i>Aug. 2022 – Jul. 2026</i>
Actively participated in departmental technical workshops, academic seminars, group discussions, and soft skills training alongside Industrial Electrical Engineering students.	
UKM Dirgantara PENS (Aeronautics & Aerospace Student Club) <i>Active Member</i>	Surabaya, Indonesia <i>Aug. 2023 – Jul. 2025</i>
Engaged in campus-wide aerospace and aeromodelling activities, technical workshops, and team-building initiatives across various engineering departments at PENS.	

PUBLICATIONS & INTELLECTUAL PROPERTY

Simple Patent Application (DGIP / PENS) <i>Directorate General of Intellectual Property</i>	Surabaya, Indonesia <i>2026</i>
Invention Title: Solar Panel Maximum Power Point Tracking (MPPT) System Based on WODE Algorithm for Submersible Water Pump Control[cite: 1, 2].	
Copyright (HAKI Kemenkumham RI) <i>Directorate General of Intellectual Property</i>	Surabaya, Indonesia <i>2024 – 2026</i>
Officially registered for the EcoMow Technical Manual (No. EC00202499819) and the Real-Time MPPT Python PyQt6 Dashboard Monitoring Software[cite: 1, 2].	

TECHNICAL SKILLS & CERTIFICATIONS

Language Proficiency: PENS English Proficiency Test (PEPT) Score: 527 (Equivalent to TOEFL ITP, Surabaya, 2026)	
CAD & Design Software: Autodesk Revit, AutoCAD, Fusion 360, Autodesk Inventor, EAGLE PCB, MATLAB/Simulink	
Programming & Data Analytics: Python (Pandas, NumPy, Matplotlib, PyQt6), SQL (PostgreSQL), Power BI, Node-RED, Microsoft Office Suite	
Power Systems & Automation: Ćuk/SEPIC Converters, MPPT Topologies, ETAP, Power Distribution, Transformer Maintenance, Switchgear, PLC Programming (Siemens TIA Portal, Omron CX-Programmer), SCADA Systems[cite: 1, 2]	
Embedded Hardware & Firmware: STM32 (CubeIDE/C++), Hardware FPU, Schematic & PCB Layout (EAGLE), Isolated Sensors (AMC1100, ACS712), Git, VS Code[cite: 1, 2]	
Standards & Compliance: PUIL, National Electric Code (NEC), NFPA, Occupational Health & Safety (K3/OHS), Lock-Out/Tag-Out (LOTO)	